Water Usage Report

Green Farm · Tub 1 · Wed 4 Feb to Tue 10 Feb
Generated 10 Feb 2026, 09:49 SAST

This week, do this

1 **Water Tub 1 today (Tue 10 Feb)**: about 36 mm (about 36 L for each square metre).

Low confidence

Why: Your soil has dried to the point where tomato starts to struggle, and no rain is expected today. (note 1)

How your farm did this week

Zone	Water given	Water needed	Verdict	Pump runs	Rain
Tub 1 (Tomato)	not known (no flow rate)	not known	Not enough data	7	0 mm

Did my crop get the water it needed?

We could not work out what the crop needed: that takes the crop, the zone's size and weather for these days.

Was my soil in the healthy range?

No chart: the soil sensor's wet/dry setup has not been done, so its readings are not moisture yet.

What's coming this week?

Wed 11 Feb	Thu 12 Feb	Fri 13 Feb	Sat 14 Feb	Sun 15 Feb	Mon 16 Feb	Tue 17 Feb
31 °C	32 °C	33 °C	34 °C	35 °C	36 °C	37 °C
Rain 0 mm	Rain 0 mm	Rain 8 mm	Rain 0 mm	Rain 0 mm	Rain 0 mm	Rain 0 mm
Water Tub 1	No watering needed	Rain covers it	No watering needed	No watering needed	No watering needed	No watering needed

Area weather vs at your plants

Your tub sensor sent no readings in this period, so only the area forecast was used.

Improve your advice

- Tub 1: Add the planting date for advice that follows your crop's growth.

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Technical appendix

For extension officers and markers. Figures and sources behind the advice; FAO-56 is Allen et al. (1998), page numbers as printed.

Tub 1

Figure	Value	Source
Soil type	Loam / garden soil	FAO-56 Table 19 (Loam): Table 19 'Loam': theta_FC 0.20-0.30, theta_WP 0.07-0.17. Midpoints used.
Root depth	0.30 m	limited by this zone's depth; FAO-56 Table 22 gives 0.7 m for tomato
Water the soil can hold	39 mm	FAO-56 Eq. 82 from Table 19 soil values and Table 22 root depth
Crop water need factor today	1.15	FAO-56 Table 12, the highest of this crop's values - assumed peak growth, as there is no planting date
Reference water use today	5.2 mm	FAO-56 Eq. 6 (Penman-Monteith), inputs from weather_app
This crop's water use today	5.9 mm	FAO-56 Eq. 58: crop factor x reference water use
Dryness it can stand before watering	14 mm	FAO-56 Eq. 83 with the depletion fraction from Table 22, adjusted for today's water use and for soil type
How dry the soil is now	36 mm short of full	FAO-56 Eq. 85 water balance, started from a full profile
TAW	39 mm	FAO-56 Eq. 82, p. 162; Table 19 p. 144; Table 22 pp. 163-165
RAW	14.1 mm	FAO-56 Eq. 83, p. 162
p (adjusted)	0.4	FAO-56 Table 22 p. 163-165; p. 162 ETc adjustment; p. 167 soil adjustment
Kc	1.2	FAO-56 Table 12 pp. 110-114, highest value (stage assumed)
Confidence	Low: based on the weather; the planting date is missing, so peak growth is assumed.	ConfidenceRules (DATA RULE)

- This zone has no planting date, so its growth stage is unknown. The crop was assumed to be at its highest water use (assumed peak growth), so the advice errs toward enough water.

- This zone has no soil type set, so garden soil (loam) was assumed. Setting the real soil type changes how much water the soil holds.

- This zone has no area recorded, so water amounts in litres cannot be worked out.

- This zone has no flow rate recorded, so watering times in minutes cannot be worked out.

- This zone's sensor has not had its wet and dry setup done, so soil readings are shown as raw numbers and cannot be turned into a moisture percentage. Watering advice still works from the weather and the crop.

- FAO-56 publishes tomato Kc_end as the range 0.70-0.90 rather than a single value. The midpoint 0.80 is used, so late-season figures carry that spread.

- Stage lengths come from FAO-56 Table 11 (Table 11, Apr/May, Mediterranean), which the book says are indicative only - a real season may run shorter or longer.

Assumptions

- Litres are estimated as pump running time x the zone's flow rate. There is no flow meter on any AgroFlow kit.

- Reference evapotranspiration (ETo) is FAO-56 Penman-Monteith (Eq. 6, p. 24) from the weather app. Solar radiation is always estimated from the day's temperature range (Eq. 50, p. 60); missing humidity is estimated from the temperatures (Eq. 64, p. 124); missing temperatures come from the tub's own sensor.

- Crop water need is ETc = Kc x ETo (Eq. 58, p. 103), with Kc from Table 12 joined by Eq. 66 (p. 132).

- The daily soil water balance is FAO-56 Eq. 85 (p. 170); stress days are days with Ks < 1 (Eq. 84, p. 169).

- Watering is advised when the root-zone depletion reaches RAW (p. 171): Dr >= RAW.

- Soil sensor readings are relative (wet/dry setup), not a measured volume of water.

- Verdicts: more than 1.2 x the need is too much, less than 0.8 x is too little (VerdictRules, DATA RULE).

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Why we said this

1. **Your soil has dried to the point where tomato starts to struggle, and no rain is expected today.**
   - Crop: Tomato (this zone's setup)
   - Crop stage: assumed peak growth (no planting date) (FAO crop water guide)
   - Soil: Loam / garden soil (assumed - not set yet) (FAO crop water guide)
   - How dry the soil is: about 36 mm short of full (weather app)
   - Watering point: about 14.1 mm short of full (FAO crop water guide)
   - Rain expected today: 0 mm (weather app)
   - Water your crop will use today: 5.9 mm (weather app and FAO crop water guide)
   - How much: about 36 mm (36 L per square metre) (FAO crop water guide)

   The rule: We suggest watering once your crop has used about 36% of the water your soil can hold for it.
   How sure: Low: based on the weather; the planting date is missing, so peak growth is assumed.
   What would change it: If 5.9 mm of rain or more falls today, you can skip this.
   Technical: Dr 36 mm, RAW 14.1 mm, TAW 39 mm, p 0.36, Kc 1.15 (assumed stage), ETo 5.2 mm, ETc 5.9 mm. FAO-56 Eq. 58 (p. 103), 82-85 (pp. 162-170). WaterRules.ShouldWaterToday (FAO-56 p. 171): Dr >= RAW.